

# TrES-5b

R-0849

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_TRES-5\_170518 · created 2026-07-28T20:30:05+00:00

BENCHMARK: ACCURATE — fit consistent with published values

## Results

|                                                   |                                |
|---------------------------------------------------|--------------------------------|
| Mid-Transit Time (Tmid)                           | 2457891.9286 +/- 0.002 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.157 +/- 0.034                |
| Transit depth (Rp/Rs)^2                           | 2.47 +/- 1.08 [%]              |
| Orbital Inclination (inc)                         | 82.82 +/- 2.32                 |
| Airmass coefficient 1 (a1)                        | 1.289 +/- 0.025                |
| Airmass coefficient 2 (a2)                        | -0.18 +/- 0.025                |
| Scatter in the residuals of the lightcurve fit is | 1.95 %                         |
| Best Comparison Star                              | None                           |
| Optimal Aperture                                  | 2.42                           |
| Optimal Annulus                                   | 11.5                           |
| Transit Duration (day)                            | 0.061 +/- 0.022                |

## Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                          |
|---------------------------|------------------------------------------|
| Rp/R■ fit vs published    | 0.157 ± 0.034 vs 0.142 (deviation 0.44σ) |
| Duration fit vs published | 0.061 d vs 0.0758 d (deviation 0.67σ)    |
| Mid-time O–C              | 5.7 min                                  |
| Depth SNR                 | 4.6                                      |
| Residual scatter          | 1.95 %                                   |

## Light curve

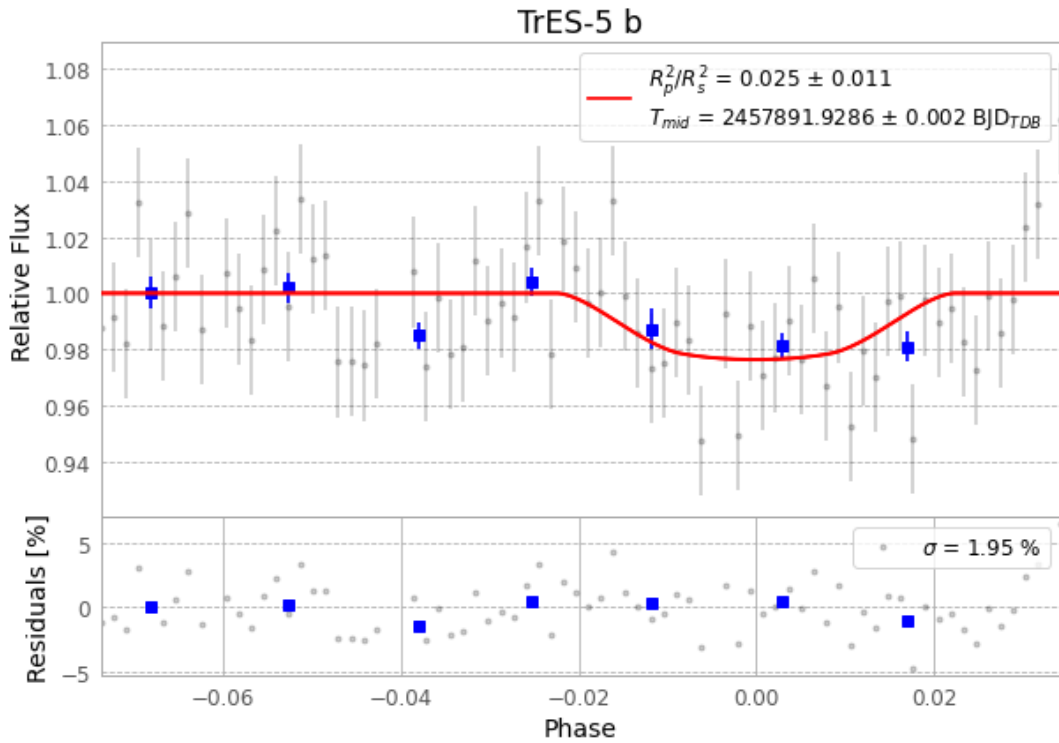


Figure 1 — FinalLightCurve\_TrES-5 b\_2017-05-18.png (SHA-256 6f1a448c21901075...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [375, 218], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 9b86945fedeb0dda...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit ead60b9

## Data provenance

75 FITS frames (49 MB), TRES-5170518073912.FITS → TRES-5170518112941.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-170518.log (e1125a3dcf73...), FinalLightCurve\_TrES-5 b\_2017-05-18.png (6f1a448c2190...), FinalLightCurve\_TrES-5 b\_2017-05-18.csv (5a83652ad3d5...)

## Compute resources

Wall time 130.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 20:30Z.